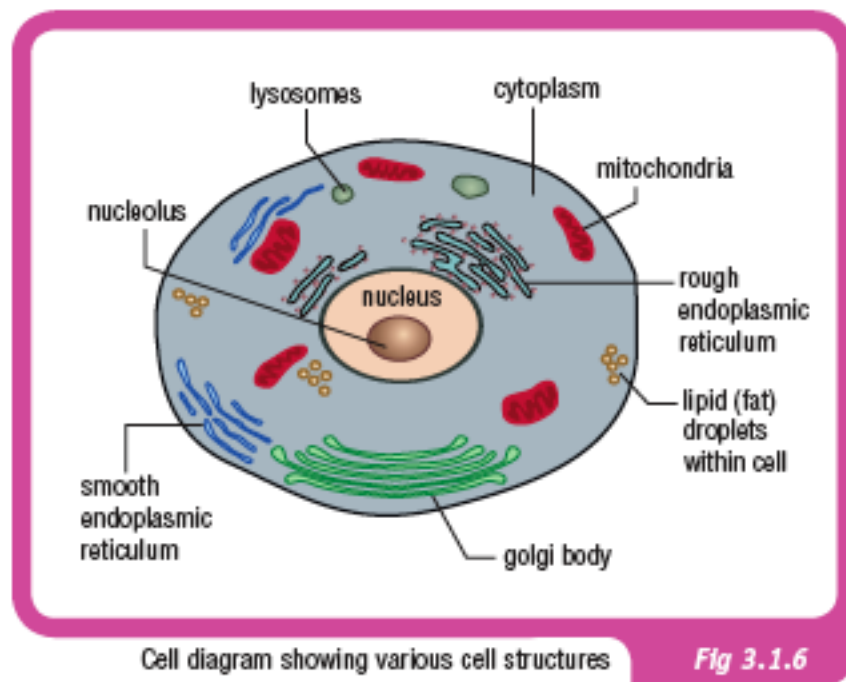


THE CELL



Functions of Organelles

- Nucleus**
- Controls all cell functions and activity.
 - Contains the genetic material (DNA).

- Cell Membrane**
- Also called the *plasma membrane*.
 - Semi-permeable (has “holes” in it) which allows the entry of food and O₂ and exit of waste products from respiration.

- Cell Wall**
- Provides strength for plant cells (made of cellulose).
 - Doesn't exist in animal cells.

- Nucleolus**
- Produces ribosomes.

- Ribosomes**
- Produce proteins for the cell to use.

- Mitochondria**
- Produce energy for the cell by respiration.

- Endoplasmic Reticulum**
- Tubular structure that transports cell products.
 - **Smooth E.R.** (without ribosomes) - transports protein for the cell.
 - **Rough E.R.** (with ribosomes) - transports protein for *export* from cell.

- Golgi Complex**
- Packages wastes for export from the cell.

- Vesicles**
- Packages produced by the Golgi complex.

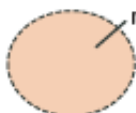
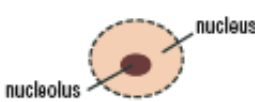
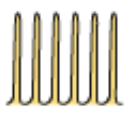
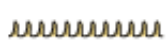
- Centrioles**
- Produce spindles for chromosome separation in cell division.

Lysosomes

- Principal digestive organelles of the cell.
- Produce digestive enzymes.

Cell organelles and their function

Fig 3.1.7

Organelle	Structure	Function
Cell membrane		A selectively permeable membrane through which substances outside the cell may be absorbed and cell products may be released. Primary functions are protection and isolation from the environment
Endoplasmic reticulum		Greatly expanded surface area for biochemical reactions, which normally occur at or across membrane surfaces. Endoplasmic reticulum may be smooth (without ribosomes—structures that are involved in protein synthesis in the cytoplasm) or rough (with ribosomes scattered across its surface)
Nucleus (chromosomes)		Contains the DNA in chromosomes and therefore the information to regulate the metabolism and reproduction of the cell. The nuclear membrane provides a selective barrier between substances in the nucleus and cytoplasm. The nuclear pores control the movement of substances
Nucleolus		The nucleolus is involved in the construction of ribosomes
Ribosomes		Sites of protein synthesis (shown as dots lining the endoplasmic reticulum). They consist of two parts—a larger piece and a smaller piece. Both are made of a complex of ribosomal RNA and protein
Golgi apparatus		Site of production of cellular secretions—the 'packaging department' of the cell. Found associated with smooth endoplasmic reticulum
Mitochondria		The 'powerhouses' of the cell responsible for energy production. Cells with a large energy requirement (as in muscle cells) have a greater number of mitochondria. Contain their own DNA, different from the chromosomal DNA
Lysosomes		Lysosomes are the principal digestive organelles of the cell. They produce powerful digestive enzymes, which aid in disposal of intracellular bacteria and other foreign bodies, as well as unwanted organelles. They may cause destruction of the cell if ruptured
Centrioles		Centrioles form poles and spindles (extensions of the microtubules) for cell division, moving the chromosomes apart during mitosis and meiosis. Capable of self-replication
Cilia		Movement of materials over a surface
Microvilli		Absorption of extracellular materials